Teach Guide — Session 1

?What Is Machine Learning

.short slides. Read in 10 minutes before class. Full script: teaching-notes.md 8

Prep checklist (night before)

- PART 0 demo cell runs fresh (Kernel → Restart & run up to it)
- Judge green on the solution notebook: `python judge/run.py notebook/s01_notebook_solution.ipynb` → 4/4
- `assets/titanic.csv` is in place (offline fallback)
- Print the fruit-table handout for the fruit game
- Lights dimmable for the hook if you can

Run of show (60 min)

Time	Segment	Where
0–5	Hook: the animal game, 4 beats	PART 0 demo cell
5–10	Rules vs learning (the two ways)	slides, fruit story
10–20	Features & labels (fruit game)	A1–A2 code-along
20–30	Supervised learning, study/exam	slides + A3
30–40	Titanic as pictures	A4–A6
40–50	THE MOMENT: `fit()` in 3 lines	A7–A9
50–55	Oral quiz	slides
55–60	Handout + Ship + judge command	B1–B4 announced

The hook — protect the 4 beats

The demo cell prints «——— مرحلهٔ N ———» before each beat. Wait for the marker, then say the line.

1. **The table** — "This is ALL the machine gets." Let them shrug.
2. **The silent fit** — one line, nothing prints. "That was the whole spell."
3. **The line** — "No human drew this." Do NOT explain how until slide 9.
4. **The quiz + reveal** — puppy wrong, elephant silly, then: the whole secret = ONE question.

HONESTY BEAT (fruit map, later): 2 of 6 rows defeat hand rules — data, not cleverness, is the point.

The moment to protect (40–50)

Cell A7–A9: `DecisionTreeClassifier` + `fit` + `predict` on 712 Titanic rows.

- Ceremony: count down, class says "LEARN", run. Under a second for 712 lives.
- Name it: "This object is a **مدل (Model)** — the student that studied."
- Let them see the tree plot (A9) — THEIR machine, not yours.

Watch for: students copying without running. Everyone runs the three lines.

Misconceptions to catch

They say	You say
"The machine understands cats"	It reads numbers in a table — nothing more
"The red line is magic"	One question it picked from the columns — Session 3 shows HOW
"More examples always fix it"	Near the line, guesses still break (the puppy)
"fit() is slow magic"	It just ran — in under a second

Judge + homework handoff

- At-home: B1–B4 in the same notebook; ~20–30 min; due before the problem-solving session.
- The judge command is ON the at-home slide: `python judge/run.py s01_notebook.ipynb`
- First run is all red — **say this out loud**, or half the class panics.
- Judge reads undefined starters ("did you fill in cell B2?") and broken files kindly.

Ship + cliffhanger + retro

Ship tonight: notebook committed to their repo — `git add . && git commit -m "S1: my first model" && git push`. Then the family demo: "I trained an AI today."

Cliffhanger (verbatim): the machine asked ONE question — *why THAT one?* Next session: the 20-questions machine.

Within 48h: fill the retro block in the session README (confused / dragged / delighted).